



Big Data Challenges in Healthcare

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Agenda

- About Me and Abt Global
- Big Data Challenges in Healthcare
- Home Health Quality Measures
- Generalized Linear Mixed Models (GLMMs)
- Databricks
- Bootstrapping and Scalability
- Advice on Preparing for Industry





About Me and Abt Global





About Me



Zachary Allen



Johnson & Johnson Data Science Intern
Jannsen Research & Development

Jan. – June '23



B.S. Computer Science, Stats Minor
• TA for Databases 1 in Fall '23

Dec. '23



Data Scientist 1 @ Abt Global

March '24 – Present





About Abt Global

- Mission-driven consulting and research firm.
- Engine for social impact in 50+ countries.
- Provide a wide variety of capabilities
 - Technical Assistance and Implementation
 - Research, Monitoring, and Evaluation
 - Digital Transformation and Data Services
 - Marketing and Communications





About Abt Global

- Primary Markets:

- Climate, Energy, and Environment
- Education & Workforce Transformation
- Governance and Justice
- Health Service Delivery and Health Systems

- Housing and Asset Building
- Workforce & Economic Mobility

- Domestic Clients



FEMA

+ many more!





Big Data Challenges in Healthcare





Benefits of Big Data in Healthcare

- Improving the quality of healthcare services
- Improving population health
- Reducing costs for patients and providers
- Supporting research activity
- Enabling real-time alerts (wearables, IoT, etc.)
- Preventing cyberattacks and enhancing security
- Managing on-site personnel (staffing)





Big Data Challenges in Healthcare

Data
Governance

Security

Vendor Lock-in

Implementation
& Maintenance

Organizational-
level

Physician-level

Interoperability

Trust

Laws, Policy & Regulations





Project Deep Dive: Home Health Quality Measures





Care Compare

- Care Compare: <https://www.medicare.gov/care-compare/>
- Enables Medicare beneficiaries to find and compare different Medicare providers based on location.





Terminology

- **Center for Medicare & Medicaid Services (CMS)**
 - Federal agency within the U.S department of Health and Human Services
- **Medicare**
 - Federal health insurance for those 65 and older (with exceptions)
- **Home Health Agencies (HHAs)**
 - Provide skilled, short-term services in-home. Typically ordered by a doctor to help with recovery following an inpatient hospital stay, rehabilitation, etc.
 - HHAs provide care, Beneficiaries receive care





Discharge to Community (DTC) Measure

- Measures are complicated fractions. For a given provider:
 - DTC Numerator:
 - Number of home health stays in which patients were discharged to the community and do not have unplanned admission to a hospital and remain alive within 31 days of the discharge.
 - DTC Denominator:
 - Number of home health stays over 2-year observation period
 - Remove claims for patients who are under 18, discharged to another HHA, etc.

Stays are series of consecutive Medicare claims (bills)

Discharged to community means one can function without home health services





Issues with DTC

- Since providers are location based, they may be **unfairly penalized** by the numerator if the population they serve is unhealthier compared to the state or national average
- We can use statistical modeling to “risk adjust”



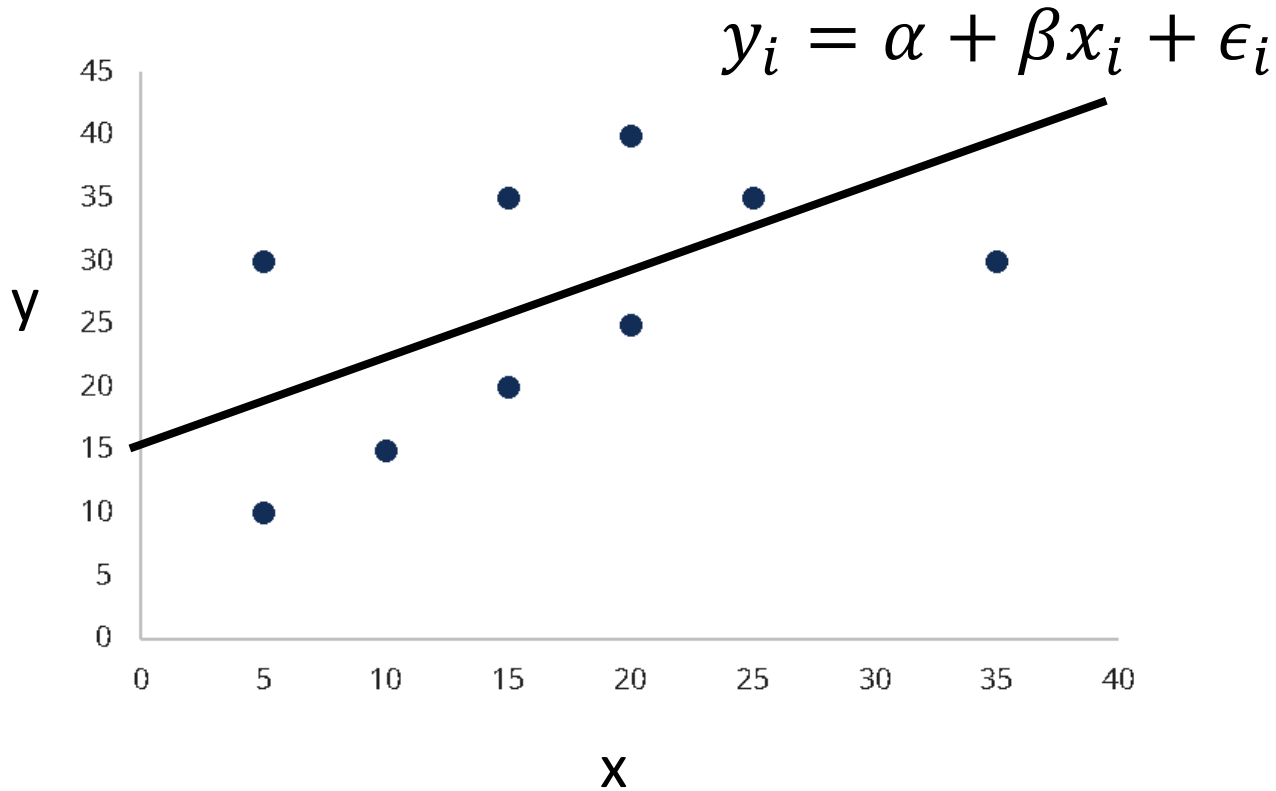


Generalized Linear Mixed Models (GLMMs)





Simple Linear Regression



- Closed-form Solution

$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

- Linear relationship between x and y





Sample Data

Response	Provider	Age > 85	High-Living Score	Prior Condition
0	A	0	1	1
1	A	1	0	0
0	B	0	0	0
1	C	0	1	1
1	C	1	1	0
0	D	1	0	1

Several hundred
other variables

Several million
rows

Tens of thousands
of providers



Binary
Response:
0 = no DTC
1 = yes DTC

Providers can
have 1+
observations



Model Definition

Fixed Effects

Ignoring Provider

$$\boxed{y} = \alpha + \beta_1 \boxed{x_1} + \beta_2 \boxed{x_2} + \beta_3 \boxed{x_3} + \dots$$

Binary response

Age > 85 High-living Score Prior Condition

Adding Provider as Fixed Effects

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_{301} \boxed{x_{301}} + \beta_{302} \boxed{x_{302}} + \dots$$

Using Provider A? Using Provider B? Tens of thousands

Issues

- y must be between 0 and 1
 - No closed-form solution (need approximation algorithm)
 - Adding parameters increases time complexity





Model Definition

Random Effect

Fixed Effects

$$g(E[y_{it}|\mu_i]) = \alpha + \mu_i + \beta_1 x_{it1} + \beta_2 x_{it2} + \beta_3 x_{it3} + \dots$$

- i is the provider number, t is the observation number for each provider
- μ_i is a Random Independent Variable that is normally distributed $N(0, \sigma^2)$
 - Not based on x (unobserved)
 - **Measures the impact a given provider has on the response**

Benefits

- We only need to estimate 1 more parameter σ^2 , the variability among providers
- Can make inferences on the entire population of providers
- Response is conditional on provider





Calculating New DTC Value

Provider	Age > 85	HLS	Prior Condition	True Response	Prediction w/ Fixed Effects	Prediction w/ Fixed and Random Effects
A	0	1	1	0	0.2	0.05
A	1	0	0	1	0.6	0.99
<u>Average:</u>				0.5	0.4	0.52

$$\text{Old DTC} \times \frac{\text{avg}(\text{predictions w/ fixed + random})}{\text{avg}(\text{predictions w/ fixed})} = \text{New DTC}$$

$$0.5 \times \frac{0.52}{0.4} = 0.65$$





Databricks and Distributed Computing





What is Spark?

- Apache Spark: open-source, multi-language engine for doing large-scale data engineering, data science, or machine learning tasks
 - Single-node machines or clusters
 - Scala, Python, SQL, or R
- Includes libraries for:
 - Machine Learning (MLlib)
 - Stream Processing (Spark Streaming)
 - Graph Processing (GraphX)

Spark: The Definitive Guide: Big Data Processing Made Simple (1st Edition) by Bill Chambers & Matei Zaharia





What is Databricks?



databricks

- Founded in 2013 ([CMU Lecture on technical history](#))
- Spark + many features (paid)
 - SAAS Platform → go to a website to use it
 - Multi-cloud (AWS, Azure, GCP)
 - **Notebook interface for Spark instances running on compute clusters**
 - Git integration with repos
 - Job creation and workflow orchestration
 - Governance & Security Layers
 - [Data Lakehouse](#)

- Direct competitor to Snowflake





Bootstrapping and Scalability





Issues with Random Effects

- Our Random Effects are single estimates for each provider. We currently have no way of measuring the uncertainty of these estimates.
- Bootstrapping:
 - Sample (with replacement) our original dataset
 - Fit the GLMM using the sample data
 - Calculate the New DTC value for each provider
 - Repeat this process 200 times

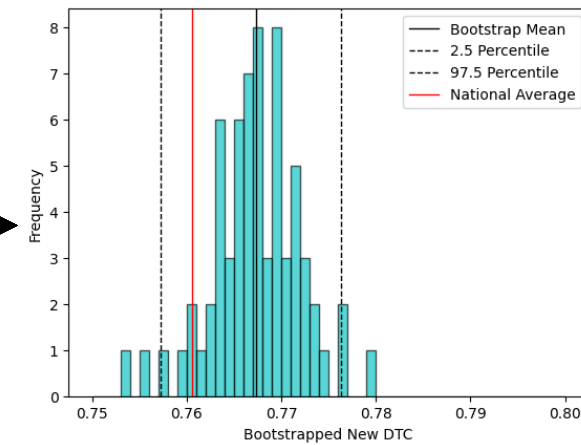
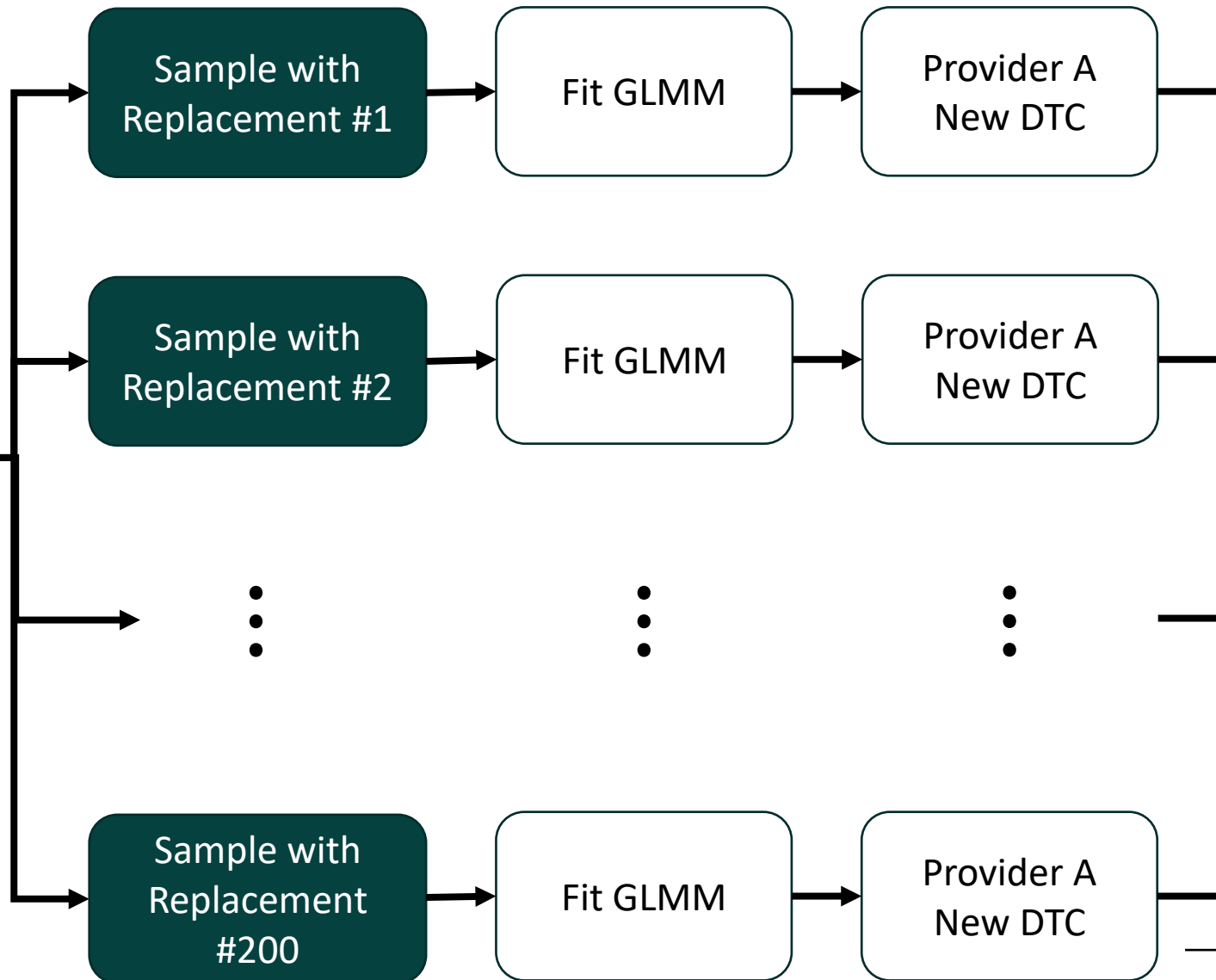




Bootstrap Visual

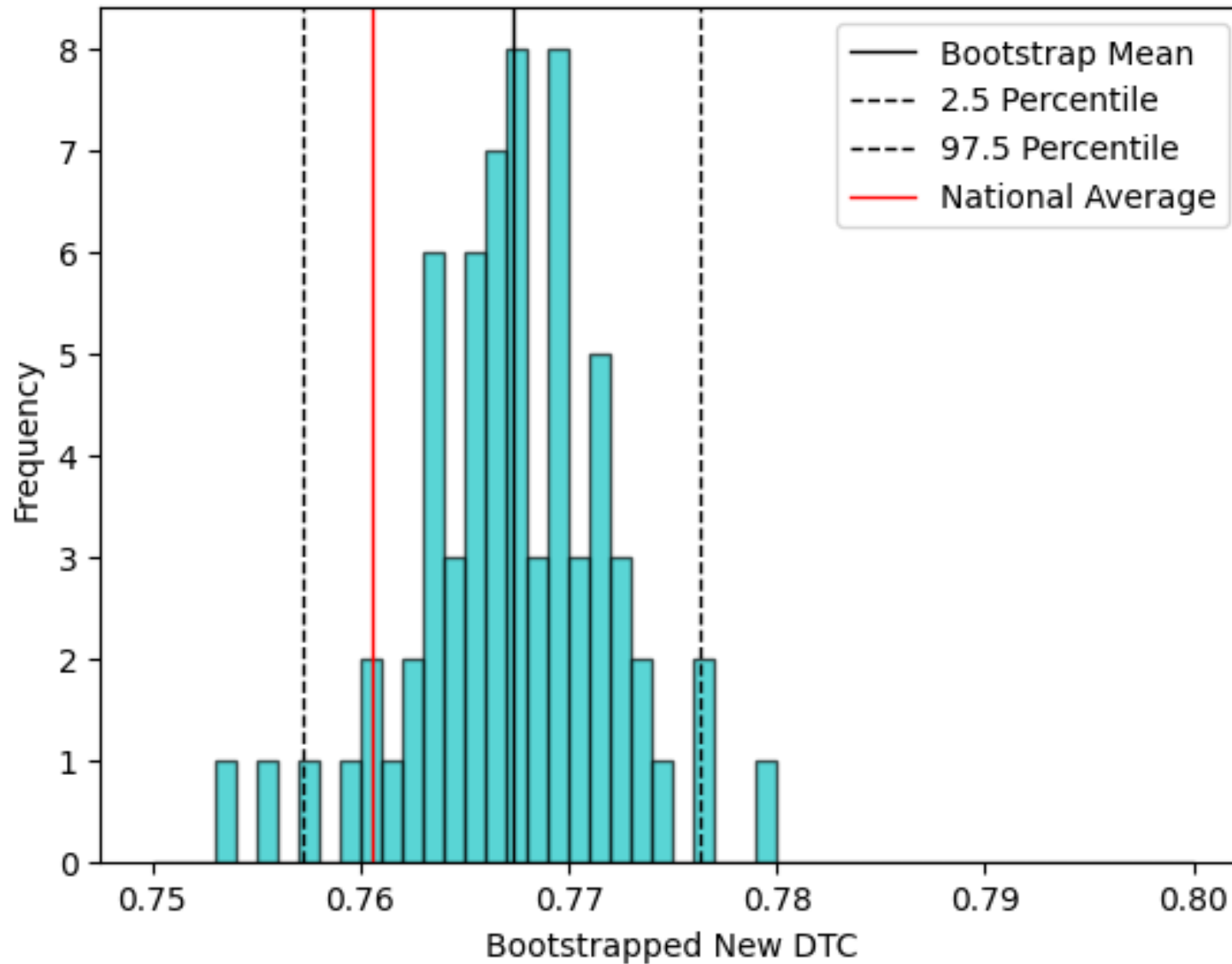
Original
Dataset

Response	Provider	Age > 85	Sex	Prior Condition
0	A	0	1	1
1	A	1	0	0
0	B	0	0	0
1	C	0	1	1
1	C	1	1	0
0	D	1	0	1





Bootstrap Visual





Running the Model on Databricks

- Use Python GPBoost Package on Databricks
 - **Issue**: Memory allocation on Compute Cluster is too small
 - Can't load in Pandas DataFrame let alone run the model
 - Can't wait for memory increase
- Photon-ml: GLM/GLMM library built for Spark
 - Made by LinkedIn
 - Old, hasn't been updated in years
 - Limited documentation, difficult to setup

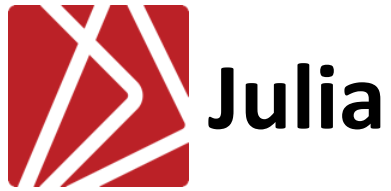




Running the Model off Databricks

- Using GPBoost on Abt Shared Compute
 - 2+ hours per iteration
 - Uses 100% memory, can't run in parallel ✖
- Using R Package lme4
 - ~2 hours per iteration
 - Still uses 100% memory, prone to crashing ✖





- High-level, high-performance programming language for numerical analysis
- Started in 2012, stable v1.0 in 2018
- Let's developer have control over memory allocation
- Designed for parallelism
- If done right, generally faster than R or Python
- Author of package for mixed models suggested 75x performance improvement





Julia Results

- Using MixedModels.jl Package on Abt
 - ~ 45 minutes per iteration
 - After optimizations, 40-45% memory usage 
 - Running code in functions, not in global scope
 - Pre-allocating large DataFrames
 - Performing Garbage Collection after each iteration





Advice on Preparing for Industry





UF Opportunities

- Courses:

- STA4241 Statistical Modeling in R or STA3180 Statistical Modeling
- Any Python-based Data Science or ML course
- CIS4301 – Intro to Databases 1

- Other

- Center for Undergraduate Research (CUR)
- UF Career Center and Career Fairs
- Teaching Assistant
- Clubs!





Resumes

- Formatting:
 - Use common, single-column, 1-page, boring templates
 - Jake's Resume (LaTeX) is a good starting point
- Content:
 - Use [XYZ](#). Accomplished X as measured by Y, by doing Z
 - Use numbers and metrics but be able to defend them!
 - Use [F-shaped reading pattern](#)
 - Have multiple people review your resume





Projects

- Group projects
 - Commonly asked interview question
- **Deploy your projects**
 - Moving from notebooks to something interactive (webpage, CLI, etc.)
 - Setting up data pipelines
 - Working in the cloud (AWS, Azure, GCP)
 - “Production-ready” code, comments, documentation
 - [Full Stack Deep Learning](#), [Data Engineering Zoomcamp](#)
- Attach to interests
 - Domains: Computer Vision, Natural Language Processing, Bioinformatics, CyberSec, Geoinformatics, Deep Learning, etc.
- ✓ • Industries: Healthcare, Finance, Sports, Manufacturing, Supply Chain, etc.



Applying and Interviews

- Broaden your search
- Identify common tools/technologies in requirements
- Don't be intimidated by "100+ applicants"
- Avoid scams and way too long take-home projects
- Prepare for online coding assessments (OA)
 - Leetcode for DS&A, DataLemur for DS/SQL
- STAR Method for Behavioral Interviews





Thank You!



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